AI Medical Chatbot

Heart Disease Prediction using Hugging Face & Google Colab

1. Problem Statement

Cardiovascular diseases are a leading cause of death globally, yet early diagnosis often requires expert radiologists or cardiologists to manually analyze medical images like ECGs, X-rays, MRIs, and CT scans. This project aims to build an AI-powered medical chatbot that can automatically analyze heart-related medical images, predict whether heart disease may be present, and provide a plain-language explanation — making preliminary cardiac screening more accessible.

2. Objective

- Upload heart-related images (ECG, X-ray, MRI, CT scan)
- Analyze images using AI (Vision Transformer - ViT)
- Predict possible heart disease based on confidence scores
- Provide medical explanations and suggestions
- Create a simple AI medical assistant with Gradio UI

3. Technologies Used

- Hugging Face Transformers — pre-trained model pipeline
- Vision Transformer (ViT) — google/vit-base-patch16-224
- Gradio — web-based UI for image upload and prediction
- Python — core programming language
- Google Colab — cloud-based execution environment
- PyTorch + TorchVision — deep learning framework
- Pillow (PIL) — image processing

4. Execution Code

Step 1 — Install Required Libraries

```
!pip install transformers gradio torch torchvision pillow -q
```

Step 2 — Import Libraries

```
from transformers import pipeline
from PIL import Image
import gradio as gr
```

Step 3 — Load Hugging Face Image Classification Model

```
classifier = pipeline(
    "image-classification",
    model="google/vit-base-patch16-224"
)
```

Step 4 — Disease Explanation Function

```
def explain_result(label):
    explanations = {
        "NORMAL": """
No major heart-related abnormalities detected.
Suggestions:
* Maintain healthy diet
* Regular exercise
* Periodic health checkups
""",
        "HEART_DISEASE": """
Possible signs of heart disease detected.
Possible Conditions:
* Arrhythmia
* Coronary artery disease
* Heart muscle abnormalities
Suggestions:
* Consult a cardiologist
* ECG/ECHO recommended
* Avoid stress and smoking
""",
        "ARRHYTHMIA": """
Irregular heartbeat pattern detected.
Suggestions:
* Immediate cardiac consultation
* Monitor heart rate regularly
"""
    }
    return explanations.get(label, "Please consult a medical professional.")
```

Step 5 — Prediction Function

```
def predict_heart_disease(image):
    img = Image.fromarray(image)
    results = classifier(img)

    top_result = results[0]
```

```
    label = top_result['label']
    score = top_result['score']

    # Map confidence score to disease category
    if score > 0.60:
        disease = "HEART_DISEASE"
    elif score > 0.40:
        disease = "ARRHYTHMIA"
    else:
        disease = "NORMAL"

    explanation = explain_result(disease)

    output = f"""
Prediction Result : {disease}
Confidence Score  : {round(score * 100, 2)} %
Explanation       :
{explanation}
"""
    return output
```

Step 6 — Create Gradio Interface

```
interface = gr.Interface(
    fn=predict_heart_disease,
    inputs=gr.Image(type="numpy"),
    outputs="text",
    title="AI Medical Chatbot - Heart Disease Prediction",
    description="""
Upload a heart-related medical image (ECG/X-ray/MRI).
The AI model predicts possible heart disease and explains the condition.
"""
)
```

Step 7 — Launch the Application

```
interface.launch(debug=True)
```

5. Steps to Run in Google Colab

Step 1: Open Google Colab (colab.research.google.com)

Step 2: Create a new notebook

Step 3: Copy and paste the complete code from Section 4

Step 4: Run all cells (Runtime > Run All)

Step 5: Upload a medical image (ECG / X-ray / MRI / CT scan)

Step 6: View the prediction result, confidence score, and explanation

6. Expected Results

The Gradio interface displays three possible outcomes depending on the confidence score returned by the ViT model:

Case 1 — Heart Disease Detected (confidence score > 60%)

```
Prediction Result : HEART_DISEASE
Confidence Score  : 78.43 %
Explanation       :
  Possible signs of heart disease detected.
  Possible Conditions:
    * Arrhythmia
    * Coronary artery disease
    * Heart muscle abnormalities
  Suggestions:
    * Consult a cardiologist
    * ECG/ECHO recommended
    * Avoid stress and smoking
```

Case 2 — Arrhythmia Detected (confidence score 40%–60%)

```
Prediction Result : ARRHYTHMIA
Confidence Score  : 52.10 %
Explanation       :
  Irregular heartbeat pattern detected.
  Suggestions:
    * Immediate cardiac consultation
    * Monitor heart rate regularly
```

Case 3 — Normal (confidence score < 40%)

```
Prediction Result : NORMAL
Confidence Score  : 28.75 %
Explanation       :
  No major heart-related abnormalities detected.
  Suggestions:
    * Maintain healthy diet
    * Regular exercise
    * Periodic health checkups
```

7. Important Note

WARNING: This project is a beginner-level demo and should NOT be used for actual medical diagnosis. The model used (google/vit-base-patch16-224) is a general-purpose Vision Transformer not trained on medical datasets. The disease classification is based purely on confidence score thresholds — this is demo logic only, not real medical intelligence. For real-world healthcare systems: train models on certified medical datasets, validate results with licensed doctors, and use certified healthcare AI systems only.

8. Future Improvements

- Train on real ECG / cardiac imaging datasets
- Custom CNN training for better medical accuracy
- Integrate BioGPT for advanced medical chatbot responses
- FastAPI backend for production deployment
- React frontend for better user experience
- MongoDB database for storing patient records
- PDF report generation per patient
- Voice assistant integration
- IoT heart sensor integration for real-time monitoring

9. Conclusion

This project demonstrates how AI and Hugging Face models can be integrated to create an intelligent medical chatbot capable of analyzing heart-related images and providing disease explanations. Using the Vision Transformer (ViT) model with a Gradio interface, it provides an accessible, beginner-friendly introduction to Medical AI, Deep Learning, and AI-powered healthcare applications. It serves as a strong foundation for building more sophisticated, clinically validated cardiac screening systems in the future.